Skills Summary

Transformations, Compositions and Inverses

Use this reference while completing your worksheet or when studying.

Skill 1 — Transforming a graph

Master form: $g(x) = a\,f(x-h)+k$. Inside changes (h) move the graph horizontally and act *opposite* to their sign; outside changes (a, k) act on the output as written, with the stretch applied before the shift.

Example: $f(x) = x^2$ and $g(x) = -2f(x-1)+3$. Find $g(4)$.
Step 1. Feed the shifted input to f first, then apply the outside factor before adding — the 3 must not be multiplied.
Step 2. $g(4) = -2f(3)+3 = -2(9)+3$.
$\Rightarrow$ $\boldsymbol{g(4) = -15}$

Try it: $f(x) = x^2$ and $g(x) = 3f(x+2)-1$. Find $g(1)$.

check: $\boldsymbol{g(1) = 26}$

Skill 2 — Composing two functions

Definition. $(f\circ g)(x) = f\big(g(x)\big)$: replace every x in the rule for f by the whole expression $g(x)$.

If a composition simplifies to x, the two functions are inverses — that is the test, not a coincidence.

Example: $f(x) = x^2+2$, $g(x) = 3x-1$. Write $(f\circ g)(x)$.
Step 1. The inner rule goes in wherever x appears in f, so square the whole binomial rather than its parts.
Step 2. $(3x-1)^2+2 = 9x^2-6x+1+2$.
$\Rightarrow$ $\boldsymbol{(f\circ g)(x) = 9x^2-6x+3}$

Try it: $f(x) = x^2-3$, $g(x) = 2x+1$. Write $(f\circ g)(x)$.

check: $\boldsymbol{(f\circ g)(x) = 4x^2+4x-2}$

(!) Watch out: $\sqrt{x^2}$ is $|x|$, not x. A square root undoes a square only on non-negative inputs, which is why so many of these problems come with a restriction attached.

Skill 3 — Finding and using an inverse

Method. Write $y = f(x)$, solve for x, and rename.

Reversal rule. $(g\circ f)^{-1} = f^{-1}\circ g^{-1}$: the last operation applied is the first one undone. A restricted domain decides the sign of any square root the inverse needs.

Example: $f(x) = 4x + 9$. Find $f^{-1}(29)$.

Step 1. The rule multiplies then adds, so the inverse subtracts then divides — reverse order, opposite operations.

Step 2. $f^{-1}(x) = \dfrac{x-9}{4}$, so $f^{-1}(29) = \dfrac{20}{4}$.

$\Rightarrow \quad \boldsymbol{f^{-1}(29) = 5}$

Try it: $f(x) = \dfrac{x+6}{3}$. Find $f^{-1}(5)$.

check: $\boldsymbol{f^{-1}(5) = 9}$